

A Water-Use Spectrum: From One ChatGPT Prompt to One Hamburger

Posted by u/lesbianspider69 • 0 points • 3 comments

On a per-action basis, these three activities span orders of magnitude in freshwater consumption:

****Lowest → Highest****

****Single ChatGPT response****

Li et al. estimate GPT-3 needs about a 500 mL bottle of water for ~10–50 medium-length replies—~0.01–0.05 L (0.0026–0.013 gal) per prompt, or roughly ****75–380 prompts per U.S. gallon****. [Making AI Less “Thirsty”](<https://arxiv.org/abs/2304.03271>). This is operational water (cooling + electricity generation).

****One hour of TV in a U.S. home****

Modern TVs draw ~50–200 W. [EnergySage](<https://www.energysage.com/electricity/house-watts/how-many-watts-does-a-tv-use/>) lists that typical range. Using NREL’s national averages for evaporative (consumptive) water loss of ****0.47 gal/kWh (thermoelectric only) to 2.0 gal/kWh (thermo + hydro mix)****, an hour of viewing (0.05–0.20 kWh) consumes about ****0.02–0.40 gallons****—grid mix and TV size drive where you land in that band. [NREL PDF](<https://docs.nrel.gov/docs/fy04osti/33905.pdf>). (Calculation is mine based on those factors.)

****One hamburger****

A standard beef hamburger (bun + toppings) carries a water footprint of about ****660 gallons**** to produce, per Water Footprint Network data compiled by WaterCalculator. [WaterCalculator](<https://watercalculator.org/footprint/foods-big-water-footprint/>).

****Sources****

- [Li et al. 2025, “Making AI Less ‘Thirsty’: Uncovering and Addressing the Secret Water Footprint of AI Models” (arXiv)](<https://arxiv.org/abs/2304.03271>)

- [Torcellini et al. 2003, *Consumptive Water Use for U.S. Power Production*, NREL]
(<https://docs.nrel.gov/docs/fy04osti/33905.pdf>)
- [EnergySage: “How many watts does a TV use?”]
(<https://www.energysage.com/electricity/house-watts/how-many-watts-does-a-tv-use/>)
- [WaterCalculator: “Food’s Big Water Footprint” (WFN data)]
(<https://watercalculator.org/footprint/foods-big-water-footprint/>)

Comments

u/[deleted] • 1 points • 2025-07-22 22:41 UTC

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u/anon876094 • 1 points • 2025-07-22 22:19 UTC

Add in one hour of gameplay of an MMO...

u/lesbianspider69 • 6 points • 2025-07-22 20:45 UTC

So I’m a pro-AI vegan. Antis can’t even begin to get on my level lol